

Module 5: Approximate Inference and Bayesian Classification

INFO 521 · Machine Learning Foundations · Week 5 · module overview

What do you do when the posterior has no closed form?

Binarise the outcome, put a logistic link on the linear model, and the conjugacy from Module 4 is gone. There is no formula for the posterior any more. This module gives you two honest ways to proceed: approximate the shape, or draw samples from it.

What the lectures do

Lecture A: Logistic Regression and Newton's Method. Squashing the line, the log-likelihood that results, and Newton-Raphson using curvature to reach the answer in a handful of steps.

Lecture B: Laplace Approximation and a Taste of Sampling. Fitting a Gaussian at the mode, when the peak is not the story, and the Metropolis trick that lets you sample without ever computing the normaliser.

Where this sits in the course

This is the week the course stops having exact answers, which is also the week it starts looking like practice. It is the busiest week on the calendar: the project midpoint lands here, and the second half of the project opens on the binarised outcome this module introduces.

Week 5 at a glance

- Homework Unit 5, approximate Bayesian inference.
- Checkpoint quiz 2.1 on the Newton-Raphson update. Passing it opens Milestone 2.1 and Gate A.
- Milestones 1.3 and 1.4 are due this week.
- The project midpoint: Part 1 is due. Part 2 begins immediately after.
- Activity loop: the week04 logistic-regression explorer. No discussion this week; it is the midpoint.
- Reading: PML sections 10.1 to 10.2 and 4.6.8; Think Bayes Chapter 19.

Watch out for

Laplace only sees the peak. It is a Gaussian fitted at the mode. Skew and multimodality are invisible to it, which is fine until it is not. Lecture B shows a case where it is not.

Plan for the midpoint before it arrives. Week 5 carries two milestones, a checkpoint, a homework unit, and the Part 1 deadline. Starting it on the Wednesday will not work.